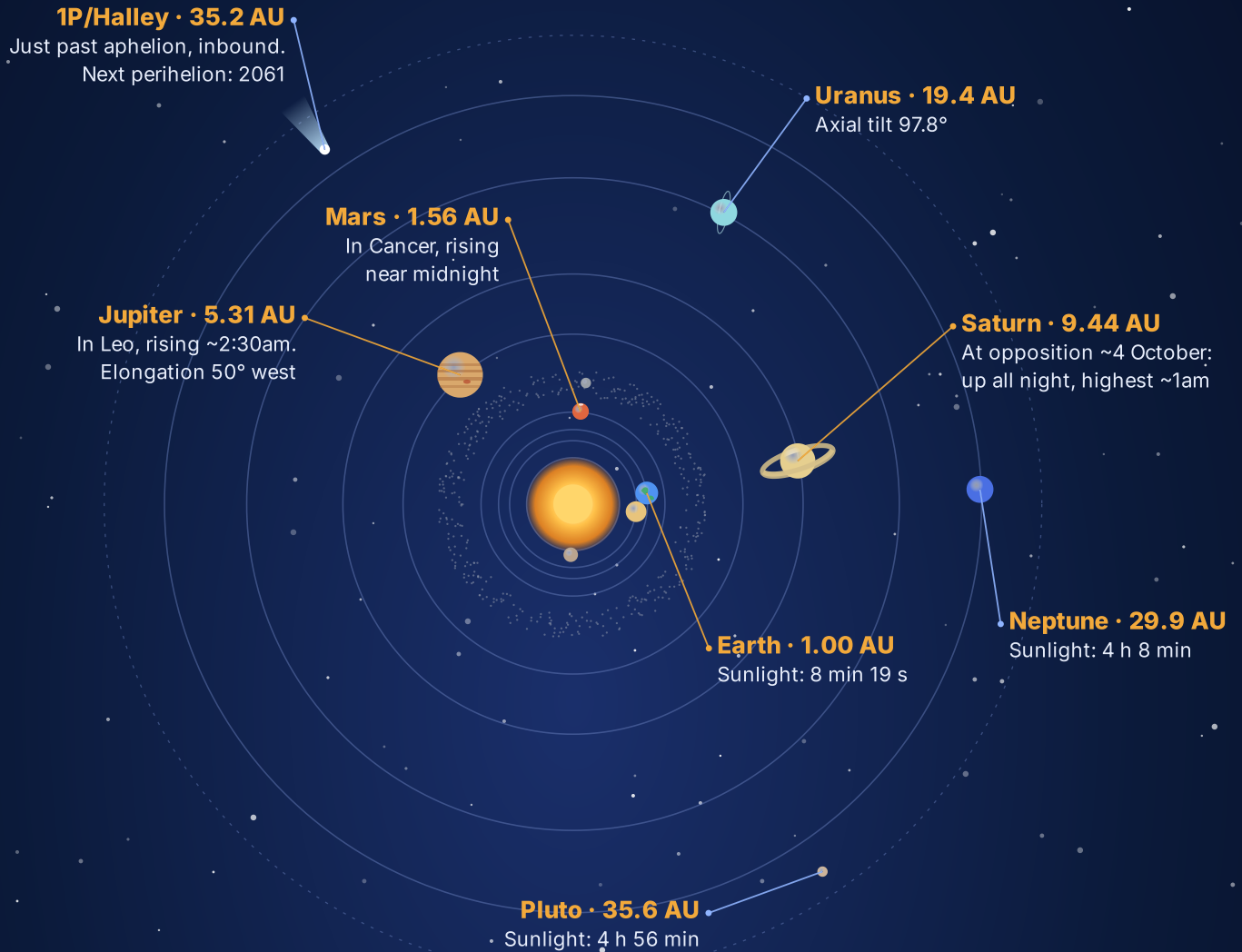


The Solar System, as live data

1.5 million objects, refreshed every night from NASA/JPL and the IAU Minor Planet Center. Explore it, query it, build with it.



Heliocentric positions on 1 October 2026, from Solar's API (ecliptic plane, square-root distance scale, bodies not to scale). Eris is at 95.5 AU, well off this map.



sol.wickedsick.com

Free · no sign-up · no adverts

Open REST API

MCP for AI assistants

Public-domain data

A field guide to the Solar System

NASA, JPL and the IAU publish a huge amount of data about the Solar System, all in the public domain. Most of it sits in formats built for professionals. Solar puts it into one clean, searchable website with a free API, so students can use real data rather than textbook tables.

1.57 m

catalogued objects

450+

known moons

2 m+close-approach
records**~900 k**

discovery records

Nightly

rebuild from source

**Every object, one page**

Orbital elements and their uncertainties, physical properties, discovery details, close approaches and sources.

**Orrery for any date**

A top-down map of the planets, stepped by day or month. Worked out from live orbital elements.

**Where is it in the sky?**

RA/Dec, constellation, altitude and azimuth, plus rise, transit and set times for your location.

**Any location format** NEW

Use your device's location, coordinates, a Google Maps link or a **what3words** address.

**Tonight's cloud cover** NEW

An hour-by-hour outlook that picks the best time to observe.

**Visibility alerts** NEW

An optional account emails you when a chosen object is up after dark.

**Open REST API**

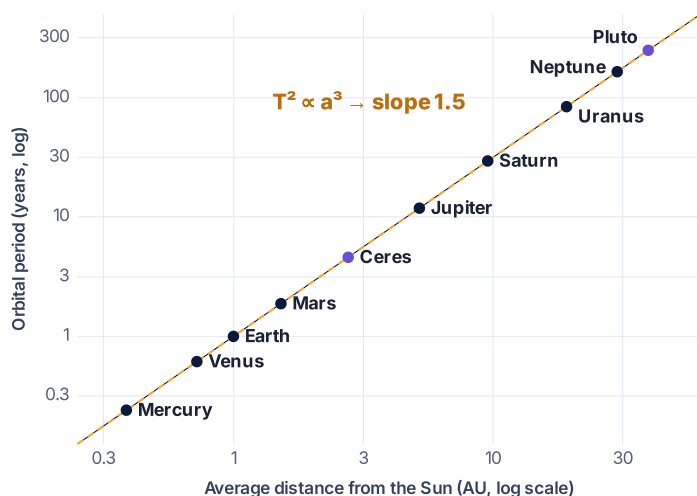
JSON endpoints with OpenAPI docs: objects, moons, close approaches, positions, sky and search.

**Download the lot**

The whole catalogue as one SQLite file (about 3.5 GB), published every night.

Kepler's third law, straight from Solar's data

Planets (navy) and dwarf planets (purple) fall on one straight line of gradient 1.5 on log-log axes.

**Honest about accuracy**

Solar works out positions with **two-body Keplerian propagation**. That's accurate to about a degree for the planets: good enough to find the right constellation and know whether something is up, but not for pointing a telescope.

Why only "about a degree"? The planets pull on each other and orbits slowly drift. It's a good discussion question at A-level.

Try it: plot Halley's Comet ($a \approx 17.8$ AU, $T \approx 76$ years). Does a comet on a stretched orbit follow the same law as the planets?

Every value links back to its source: JPL's Small-Body Database, the JPL satellite tables, NASA fact sheets and the IAU Minor Planet Center.

Real data for real lessons

Each idea uses live values from the site or API, so students work with the numbers scientists use.

Key Stage 3 Years 7–9

Weight on other worlds

Use $W = mg$ to find your weight on each planet. Solar gives $g = GM/R^2$. Why is NASA's figure for Jupiter lower?

Physics · gravity

Scale it outside

Use the distances in AU to build a scale model across the school field. Where do the asteroid belt and Halley's Comet go?

Physics · space Maths · ratio

Tilt and seasons

Compare axial tilts: Earth is 23.4°, Uranus is 97.8°. What would seasons be like on Uranus?

Physics · seasons

Key Stage 4 GCSE · Years 10–11

Orbital speed

Work out $v = 2\pi a / T$ for each planet from its distance and orbital period. Plot it against distance. Why do outer planets move more slowly?

GCSE Physics · space physics

Standard form for real

Masses, radii and distances in standard form: Earth is 5.97×10^{24} kg. Compare, convert and estimate densities.

GCSE Maths · number

Your first API

Call `/planets/Saturn moons`, read the JSON and sort the moons by size in Python.

GCSE Computer Science

Key Stage 5 A-level · Years 12–13

Weigh the Sun

Fit $T^2 = (4\pi^2/GM)a^3$ to the planets' orbits and solve for the Sun's mass. Then do the same for Jupiter using its moons.

A-level Physics · gravitational fields

Near-Earth asteroids

Filter asteroids by how close their orbits come to Earth and by size. Look at how their sizes are spread, then discuss risk and how uncertain the measurements are.

A-level Maths · statistics

SQL on 1.5 m rows

Download the SQLite catalogue. It's a real dataset for writing queries, joining tables and making indexes, and works well for NEA projects and EPQs.

A-level CS · NEA EPQ

STEM club and project ideas

"What's up tonight?" display

A Raspberry Pi and small screen in reception that shows which planets are visible, using the `/sky` endpoint.

Ask the Solar System

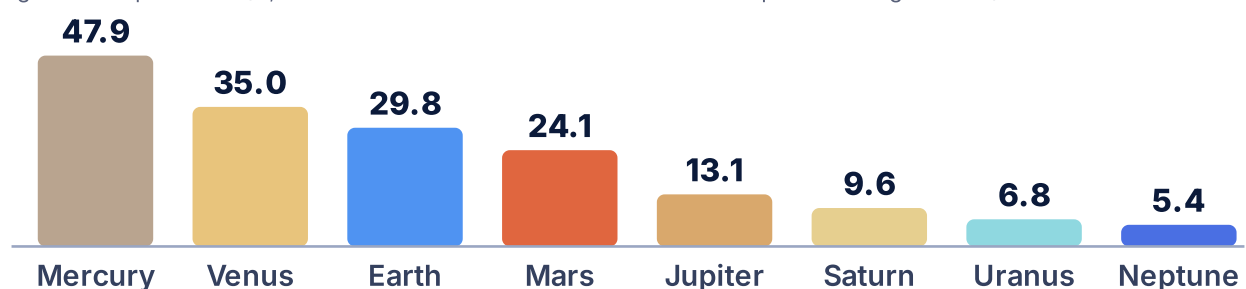
Connect an AI assistant to Solar through MCP and test its answers against the data. When is it right, and when is it wrong?

Close-approach watch

Track which asteroids pass Earth this month, how close they come, and how well their orbits are known.

Orbital speed drops with distance

Average orbital speed in km/s, worked out from Solar's distances and orbital periods using $v = 2\pi a / T$.



For stargazers, makers and developers

Solar is free for everyone, whether you're a family in the garden, an astronomy society or someone building the next app.



Stargazers and families

Enter a **what3words** address for a dark-sky spot, check the cloud outlook, and set an alert for when a planet is up after dark.



Astronomy societies

Rise, transit and set times, constellations and elongations for planets, comets and asteroids. Good for planning an outreach night.



Writers and worldbuilders

Real orbits, moons, densities and discovery stories for every known world. Your science fiction can be scientifically accurate.



AI assistants

Point Claude or another MCP-aware assistant at Solar and it can answer space questions from real data, not guesswork.

</> For developers: free, open and documented

The REST API base is `api.sol.wickedsick.com/api/v1`. Setup guides are at `sol.wickedsick.com/api`.

```
# Where is Jupiter tonight, from London?
curl "https://api.sol.wickedsick.com/api/v1/sky/Jupiter?lat=51.5&lon=-0.13"
```

MCP (streamable HTTP) for AI clients. The code is MIT-licensed on GitHub.

```
"solar-system-db": {
  "url": "https://api.sol.wickedsick.com/mcp"
}
```



Safe and simple for schools

- **Free** with no adverts. Students don't need an account.
- Location is **optional**, rounded to about 1 km, and never needed to browse.
- what3words addresses are converted, then thrown away, never stored.
- All data is **public domain** or free to reuse with credit.

⚠ **Never look at the Sun directly or through any optical instrument without a certified solar filter.**

Shape what we build next



Solar is a free community project from **Wicked Sick**, a UK technology company. We add new features all the time. Coming soon:

☀ Meteor-shower calendar

🔗 Exoplanet catalogue

☆ 3D galaxy explorer

🛰 Artificial satellites

Tell us which feature, dataset or endpoint would make the biggest difference in your department.

✉ **hello@wickedsick.com**



Why we built Solar

This data belongs to everyone, but it was hard for anyone except professionals to use. We built Solar to make it easy for students, teachers and anyone curious. It's one of our community projects, and it will stay free.